`insertClient` implementation—for example, to call a web service instead of the DB—you also have to change the tests. Not ideal.

In the following sections, you use the `akka-testkit` module to test some basic scenarios. You learn the main options to test the specific sync and async contexts.

NOTE The source code for this chapter is available at www.manning.com/books/akka-in-action-second-edition or https://github.com/franciscolopezsancho/akka-topics/tree/main/chapter04. You can find the contents of any snippet or listing in the .scala file with the same name as the class, object, or trait. For example, you can find the snippet with `object SimplifiedWorker` in the SimplifiedWorker.scala file.

4.2 Sync testing

Sync testing is intended to test an actor in isolation to check its internal logic in a white-box style. It is sequential and therefore not suitable for testing asynchronous calls that are part of the actor's logic or for testing interaction with the real implementation with other actors. Timers and the scheduler also cannot be tested this way. Using this toolkit, you can test three things: effects, reception of messages, and logs.

4.2.1 Effects

Actor *effects* are most of the methods that can be executed in an actor via the `ActorContext`. For example, the actor has been